WORKED SOLUTION

STEP I 1993, Question 7

Pure • Roots of Cubics

STEP I 1993, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q7

The key idea

A cubic with positive leading coefficient rises from $-\infty$ to $+\infty$, so everything about its roots is decided by its two turning points. If the curve's local maximum is above the axis and its local minimum is below, the graph must cross three times; otherwise it crosses fewer times. For $x^3 + ax^2 + b$ the turning points are easy to find exactly, because the derivative factorises.

Solution

The turning points

For $\mathrm{f}(x) = x^3 + Ax^2 + B$,

$$\mathrm{f}'(x) = 3x^2 + 2Ax = x(3x + 2A),$$

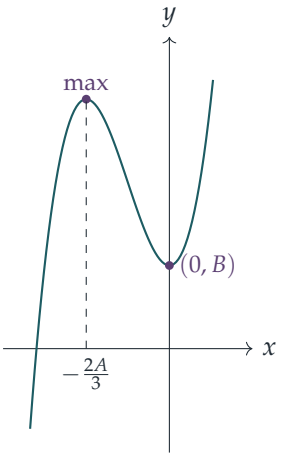
so the stationary points are at $x = 0$ and $x = -\frac{2A}{3}$. The values there are $\mathrm{f}(0) = B$ and

$$\mathrm{f}\left(-\tfrac{2A}{3}\right) = -\frac{8A^3}{27} + \frac{4A^3}{9} + B = \frac{4A^3}{27} + B.$$

Also $\mathrm{f}''(x) = 6x + 2A$, so $\mathrm{f}''(0) = 2A$ and $\mathrm{f}''(-\frac{2A}{3}) = -2A$. When $A > 0$ the point at $x = 0$ is a minimum and the one at $x = -\frac{2A}{3}$ a maximum; when $A < 0$ the roles swap.

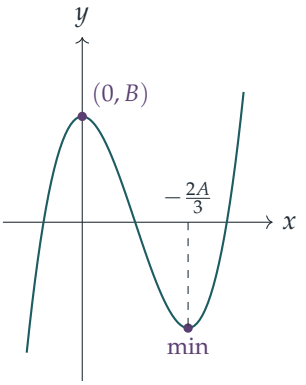
Sketch with $A > 0$, $B > 0$

The maximum is at $(-\frac{2A}{3},\, B + \frac{4A^3}{27})$, to the left of the y-axis and higher than B; the minimum is at $(0, B)$, on the positive y-axis. Both turning points are above the x-axis, so the curve crosses it exactly once, to the left of the maximum. (The figure uses $A = 3$, $B = 2$: maximum $(-2, 6)$, minimum $(0, 2)$.)



Sketch with $A < 0, B > 0$

Now the maximum is at $(0, B)$ and the minimum is at $\left(-\frac{2A}{3}, B + \frac{4A^3}{27}\right)$, to the right of the y-axis. Since $A^3 < 0$, the minimum value is less than B, and it may be positive, zero or negative. There is always one root with $x < 0$, because $\mathrm{f}(0) = B > 0$ and $\mathrm{f} \to -\infty$ as $x \to -\infty$. If the minimum dips below the axis, there are two further roots with $x > 0$ (as in the figure, $A = -3$, $B = 2$: minimum $(2, -2)$, roots $1 - \sqrt{3}$, 1, $1 + \sqrt{3}$). If the minimum is on the axis, the curve touches it there. If the minimum is above the axis, the negative root is the only one.



Counting the roots

Now take $\mathrm{g}(x) = x^3 + ax^2 + b$. If $a = 0$, then g is strictly increasing (its derivative $3x^2$ vanishes only at 0), so it has exactly one real root. So suppose $a \neq 0$. From the work above, g has two distinct turning points, with values b and $b + \frac{4a^3}{27}$.

Ask yourself How do the two turning-point values tell us the number of roots?

Write $x_1 < x_2$ for the turning points. The cubic is strictly increasing on $(-\infty, x_1]$, strictly decreasing on $[x_1, x_2]$ and strictly increasing on $[x_2, \infty)$, so it has at most one root on each piece. If $g(x_1) > 0 > g(x_2)$, the intermediate value theorem gives a root in each of $(-\infty, x_1)$, (x_1, x_2) and (x_2, ∞) (using $g \to \mp\infty$ as $x \to \mp\infty$), so there are three distinct roots. If instead both values have the same sign, or either one is zero, at least one piece has no root in its interior and the roots number at most two. The maximum is always the larger of the two values, so "one positive, one negative" is the same as "their product is negative". Hence g has three distinct real roots exactly when

$$b\left(b + \frac{4a^3}{27}\right) < 0 \quad \Longleftrightarrow \quad 27b^2 + 4a^3 b < 0.$$

(Multiplying by $27 > 0$ keeps the inequality's direction.) This also covers $a = 0$, where $27b^2 + 4a^3 b = 27b^2 \geq 0$.

The first condition

Suppose $27b^2 + 3a^3 b < 0$. Then $3a^3 b < -27b^2 \leq 0$, so $a^3 b < 0$, and therefore

$$27b^2 + 4a^3 b = \left(27b^2 + 3a^3 b\right) + a^3 b < 0 + 0 = 0.$$

By the criterion above, the equation has three distinct real roots, as required.

The second condition

As printed, the second condition repeats the inequality $27b^2 + 4a^3 b < 0$, and with "$<$" the claim is false: $a = -3$, $b = 1$ gives $27 - 108 = -81 < 0$, yet $x^3 - 3x^2 + 1$ takes the values 1 at $x = 0$ and -3 at $x = 2$, so it has three distinct real roots. The intended inequality must be the reverse one. The criterion shows that if

$$27b^2 + 4a^3 b > 0$$

(indeed if $27b^2 + 4a^3 b \geq 0$), then the equation has fewer than three distinct real roots. This completes the question.

So the exact dividing line is $27b^2 + 4a^3 b = 0$. The condition with $3a^3$ is sufficient but not necessary for three roots: $a = -2$, $b = 1$ gives $27b^2 + 3a^3 b = 3 > 0$ but $27b^2 + 4a^3 b = -5 < 0$, and $x^3 - 2x^2 + 1 = (x - 1)(x^2 - x - 1)$ has the three roots 1 and $\frac{1 \pm \sqrt{5}}{2}$.

Checks

With $A = -3$, $B = 2$ we get $x^3 - 3x^2 + 2 = (x - 1)(x^2 - 2x - 2)$, whose roots $1, 1 \pm \sqrt{3}$ agree with the sketch, and $27 \cdot 4 + 4(-27)(2) = -108 < 0$, as the criterion predicts. Numerically, for about 80,000 random rational pairs (a, b) satisfying the first condition, the stationary values always had opposite signs. Over 200,000 pairs, "three distinct roots" matched $27b^2 + 4a^3 b < 0$ exactly.

Common slips

- Forgetting the case $a = 0$, where the two turning points merge into a point of inflection.
- Treating a zero turning-point value as giving three roots: it gives a repeated root, so only two distinct ones.
- Trying to prove the second statement exactly as printed instead of noticing that it contradicts the criterion.

Beyond STEP

The quantity $-(4a^3b + 27b^2)$ is the *discriminant* of $x^3 + ax^2 + b$, the cubic analogue of $b^2 - 4ac$. It equals the product of the squared differences of the roots, so it is positive for three distinct real roots, zero for a repeated root and negative when two roots are complex. Discriminants and the symmetric functions of roots are developed in Galois theory; Allenby's *Rings, Fields and Groups* is a good introduction.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.